Do Now · retrieval

Lesson 074 · Big O: Constant, Linear and Logarithmic

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What is an algorithm's input size n?

Q2

What is a trace table?

Q3

What is an invariant?

Q4

A sorted search set of 128 candidates is halved repeatedly to one. How many halvings? Show the sequence.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Observe — Big O describes the upper bound on how work grows with input size, not the precise milliseconds of an execution.

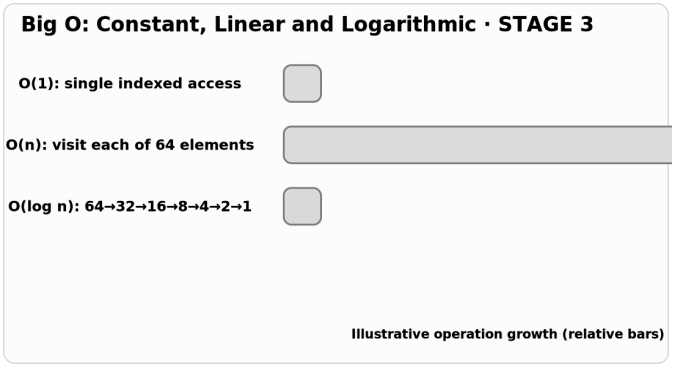
Core idea 2

Explain — A fixed direct index operation is O(1), scanning n values is O(n), and repeatedly halving n is O(log n).

Core idea 3

Evaluate — For n=64, a linear scan may inspect 64 items, while halving 64→1 takes six halvings. Sorting requirements and constants still matter.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

O(1) — Constant asymptotic work.

O(n) — Linear growth.

O(log n) — Logarithmic growth.

asymptotic — Behaviour as size becomes large.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

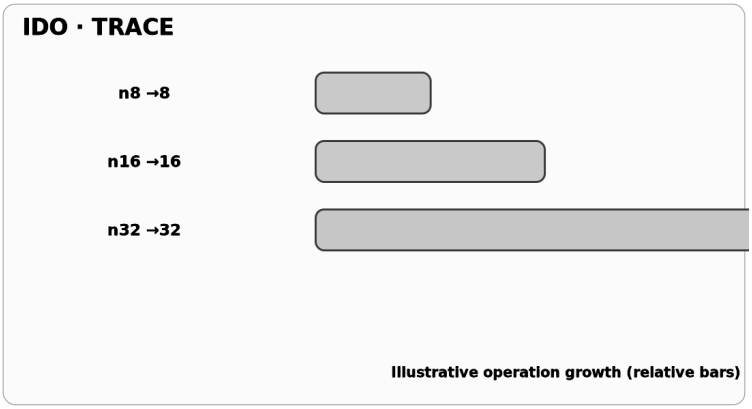
Problem

Classify loop performing one action for each item in n-element array.

Follow the worked method

1. Explain — The number of operations doubles when n doubles.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Explain
Each iteration cuts remaining size in half.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Match direct array access, full array scan and repeated halving with Big O. [6]

Question 2

For input size64, compare scan steps64 and repeated halving to1; calculate halvings. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Explain why O(log n) binary search may be unsuitable for unsorted data even if asymptotically faster than linear search. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Big O directly tells you the elapsed seconds of a program.

Use key vocabulary from memory

1. Explain O(1) in your own words.

2. Explain O(n) in your own words.

3. Explain O(log n) in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.